Contents

Edexcel IAL Pure Mathematics 2 (WMA12/01) — May 2024 Paper Review

Recall & Jargon Pack

Tutorial: Harry Williams × Phuc Nguyen · recorded 10 July 2026 · paper reviewed = **WMA12/01, Tuesday 14 May 2024** (morning, 1 h 30 m, 75 marks, 10 questions).

This pack is a fast-recall companion to the full revision guide (`guide/harry_2026-07-10_p2_may2024_revision_guide.pdf`). It compresses every term, formula and method touched in the lesson into paste-ready tables, and it is built to keep the **evidence boundaries visible at every line** — you should never have to guess whether a claim comes from the official paper or from the video.

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0. How to read this pack — provenance & status boundaries

0.1 Where each fact comes from (authority order)

Source	Role in this pack	Authority
`sources/WMA12_May2024_QP` (.txt/.pdf)	Exact question wording, marks per part	**Authoritative** — question text
`sources/WMA12_May2024_MS` (.txt/.pdf)	Correct answers, method/accuracy marks, required conditions	**Authoritative** — answers/values
`analysis/gemini_video_evidence.md`, `analysis/segment_01–04.md`	Harry's method, Phuc's attempts, exam-technique advice, timings	**Primary (lesson)** — paraphrase, not verbatim
`sources/study_log_2026-07-10.md`	Homework, next steps, logged weaknesses	Primary (admin)
`sources/loom_summary.txt`, `loom_chapters.txt`	Lesson map / chapter timeline	Secondary orientation

Source	Role in this pack	Authority
`sources/savemyexams_*.html`	Topic sanity cross-check	Cross-check only — never overrides
`shared/Edexcel_IAL_Mathematics_2018_-specification.txt`	Confirms every topic is in P2 scope	Scope truth

Rule applied throughout (QP/MS vs Harry-video): where the Gemini video ledger's transcribed *wording or numbers* disagree with the official QP/MS, **the QP/MS win.** The video is used only for *method, diagnosis of Phuc's work, and exam advice*. Every such correction is listed in §7.

> ⚠ **No verbatim transcript exists.** The Loom auto-transcript was unavailable (same-day upload). All "Harry said…" content is **Gemini 2.5 audio+visual paraphrase** — treat quotation marks as *reconstructed*, not literal.

0.2 Status legend (used in the "Lesson status" column of §6)

Tag	Meaning
Solved	Phuc got it right in the exam; lesson confirmed / polished it
Partly	Method marks earned, one mark lost — fixable slip
Taught	Harry built the method from scratch in the lesson
Blank / deferred	Left blank in the exam → homework; answers here are **MS reference only**
Setup only	Lesson reached the set-up but not the answer → MS reference only
Not reached / Not discussed	Recording ended, or part never came up → **MS reference only**

0.3 Source-limited Gemini claims (labelled, not asserted as fact)

Two lesson claims rest on a **single, internally-shaky Gemini reconstruction**. They are included because they may be real, but they are **flagged, not trusted**:

- **G1 — Q1(a) "$\frac{3}{8}x^2$".** `segment_01.md` (L77, L198, L282, L419) says Phuc wrote `3/8 x²` for the third term *and* that "the calculation was correct" — the two claims contradict. **Do not** infer a confirmed extra mark loss. The reliable takeaway is only the presentation note: write `x²`, not `+1x²`.
- **G2 — Q9(b) "k inside the integrand".** `segment_03.md` reports Phuc wrote k in place of x inside the R_2 integrand. Plausible and consistent with the surrounding teaching, but **single-source Gemini reconstruction** — labelled as such wherever it appears.

Everywhere below, cells that depend on G1/G2 are marked **[Gemini — source-limited]**, and cells whose numbers come only from the mark scheme (not the lesson) are marked **[MS-only]**.

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1. Recall tables — terms & jargon by topic

Columns are fixed: **Term · Student-friendly meaning · Exact mathematical meaning · Symbols/conditions · What it is NOT · Lesson example · Phuc/common trap.** Every "Lesson example" is anchored to this May 2024 paper; every "Phuc/common trap" is from the lesson diagnostic unless marked [MS-only].

1.1 Binomial expansion (Q1)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Binomial expansion	Multiply out a bracket to a power, term by term	$(1+u)^n = 1 + nu + \frac{n(n-1)}{2!}u^2 + \frac{n(n-1)(n-2)}{3!}u^3 + \cdots$	Real n; converges for $\|u\| < 1$ if $n \notin \mathbb{Z}^{\geq 0}$ (here $n = 9$, terminates)	Not just "$1 + nu$"; not the same as factorising	Q1(a): $\left(1 - \frac{x}{6}\right)^9 = 1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3 + \cdots$	"$+1x^2$" written where just "x^2" is wanted (MS accepts $1x^2$)
"First four terms, ascending powers"	Give the four lowest-power terms, smallest power first	Terms in x^0, x^1, x^2, x^3 in that order, simplified	Must be *simplified*; keep order low→high	Not the *whole* expansion; not descending powers	Q1(a), 3 marks	Writing an unnecessary coefficient on x^2
"Coefficient of x^m"	The number multiplying x^m after everything is collected	Sum of every product that lands on x^m	Collect **all** cross-terms reaching x^m; you may skip terms you don't need	Not a single term — often a **sum** of contributions	Q1(b): coeff of x^3 in $(10x+3)\left(1 - \frac{x}{6}\right)^9 = \frac{53}{6}$	**Missed the $10x \times x^2$ product**; took only $3 \times x^3$ → lost the final mark
Reconstruction note	—	—	—	—	Gemini also reports a written "$\frac{3}{8}x^2$" third term	**[Gemini — source-limited]** contradicts "calc correct"; not a confirmed loss

1.2 Arithmetic sequences & series (Q2)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Arithmetic sequence	Add the same amount each step	$u_n = a + (n-1)d$	$a =$ first term, $d =$ common difference, $n \in \mathbb{Z}^+$	Not geometric (that *multiplies*)	Q2: 6th term $a + 5d = 2$	Confusing the *term* formula with the *sum* formula
Common difference d	The fixed gap between consecutive terms	$d = u_{n+1} - u_n$ (constant)	Can be negative or (here) positive	Not a ratio	Q2(a): $d = 20$	—
Sum of n terms S_n	Add up the first n terms	$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a+l)$	$l =$ last term; n a positive integer	Not the nth term u_n	Q2(a): $S_{10} = -80 \Rightarrow 2a + 9d = -16$ → $a = -98,\ d = 20$	Mixing up S_n and u_n
"Smallest n such that $S_n > k$"	The first whole number of terms making the sum exceed k	Solve $S_n > k$ (a quadratic in n), take smallest integer above the positive root	Reject negative/irrelevant root; test integers each side	Not "solve $S_n = k$" — it's an inequality	Q2(b): $5n^2 - 54n - 4000 > 0 \Rightarrow n = 35$	Arithmetic slip forming/simplifying the quadratic → lost the accuracy mark (method marks banked)

1.3 Geometric sequences & exponential models (Q10)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Geometric sequence	Multiply by the same factor each step	$u_n = ar^{n-1}$	$a =$ first term, $r =$ common ratio ($r \neq 0$)	Not arithmetic (that *adds* d)	Q10 dormice: $\times 1.03$ each year	Using $n-1$ vs t indexing wrongly (see below)
Exponential growth model	"Grows $r\%$ a year" → repeated multiply	value after t yr $= (\text{start}) \times \left(1 + \frac{r}{100}\right)^t$	t measured **from the stated start** ($t = 0$ = start)	Not $\times(1 + \frac{r}{100})^{t-1}$ when t is "years *after* start"	Q10(a): $2000 \times 1.03^6 = 2390$ (3 s.f.)	Wrote $2000 \times 1.03^{7-1}$ — **"n vs t" confusion**; also rounded to 3 d.p. not 3 s.f.
Fitting $N = ab^t$	Find the start a and ratio b from two data points	Two equations → $b^{\Delta t} = \frac{N_2}{N_1}$, then back-substitute for a	Give a, b to stated s.f.; $b < 1$ = decay	Not $N = A + B(1.2)^t$ (that's not what the paper asks)	Q10(b) **[Setup only / MS-only]**: $b = 0.98,\ a \approx 4000$	Recording only reached the set-up ($t = 4, N = 3690$)
"When are they equal?"	Find the time both models give the same number	Solve model$_1$ = model$_2$ for t	Take logs; answer to stated precision	Not a guess — solve algebraically	Q10(c) **[Not reached / MS-only]**: $T = 13.9$ (1 d.p.)	Recording ended before this part

1.4 Logarithms (Q3)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Logarithm	"What power gives this number?"	$\log_b X = c \iff b^c = X$	$b > 0,\ b \neq 1,\ X > 0$	Not multiplication; $\log(X+Y) \neq \log X + \log Y$	Q3(i): base-2 logs	—
Power law	A power inside comes out front	$p \log_b X = \log_b X^p$	$X > 0$	Not $(\log_b X)^p$	$2\log_2(2-x) = \log_2(2-x)^2$	—
Product/quotient laws	Add logs = multiply; subtract = divide	$\log_b X + \log_b Y = \log_b XY$; $\log_b X - \log_b Y = \log_b \frac{X}{Y}$	Same base; args > 0	Not $\log_b(X+Y)$	$4 + \log_2(x+10) = \log_2(16(x+10))$	—
Change of base	Rewrite a log in another base	$\log_b X = \dfrac{\log_a X}{\log_a b}$	Any valid base a	—	Q3(ii): $\log_{\sqrt{a}}(a^6) = \frac{6}{1/2} = 12$	QP text-extraction blurred the base; **MS answer** $= 12$ confirms base $\sqrt{a}$
Domain / extraneous root	Some algebra answers break the "must be positive" rule — throw them out	Every log argument must stay > 0; reject roots that violate it	Check **after** solving	Not "keep both roots"	Q3(i): $x^2 - 20x - 156 = 0 \Rightarrow x = -6$ or 26; **reject** 26 (needs $2 - x > 0$) → $x = -6$	**Praised** — Phuc correctly rejected the extraneous root

1.5 Factor & remainder theorems, polynomials (Q4)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Remainder theorem	To get the remainder, just plug the root in	Dividing $f(x)$ by $(x-a)$ leaves remainder $f(a)$	From $f(x)=(x-a)q(x)+r \Rightarrow f(a)=r$	Not long division (that's the slow way)	Q4(a): $f(2)=0+21=21$	Started long division / asked "what is the remainder theorem?" — Harry taught it from scratch
Factor theorem	If it's a factor, plugging the root gives zero	$(px-q)$ a factor $\iff f\left(\frac{q}{p}\right)=0$	$r=0$ special case of remainder theorem	Not "remainder $\neq 0$"	Q4(b): $(2x-1)$ factor $\Rightarrow f(\frac{1}{2})=0 \Rightarrow k=11$	—
Command word "hence"	Use what you just proved	Reuse the previous result ($k=11$ and the known factor)	—	Not "start over"	Q4(c): use $k=11$ + factor $(2x-1)$	—
Fully factorise	Break into factors that can't go further	$2x^3+x^2+x-1=(2x-1)(x^2+x+1)$	Quadratic factor may be irreducible	Not "leave a quadratic you *could* split" — but do check first	Q4(c)(i)	Tried to factorise x^2+x+1 further and got stuck
Discriminant	Quick test for how many real roots / whether it factorises	$\Delta=b^2-4ac$: >0 two, $=0$ one, <0 none; perfect square ⇒ factorises over rationals	Real a,b,c	Not the roots themselves	Q4(c)(ii): x^2+x+1 has $\Delta=1-4=-3<0$	**Must state** $b^2-4ac=-3<0$ as the reason — MS gives A0 without it

1.6 Circle geometry (Q7)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Completing the square (circle)	Turn the messy equation into centre–radius form	$x^2+y^2+2gx+2fy+c=0 \to (x+g)^2+(y+f)^2=g^2+f^2-c$	RHS $=r^2>0$	Not the general quadratic-formula CTS	Q7(a): $x^2+y^2+8x-10y=29 \to (x+4)^2+(y-5)^2=70$	—
Centre & radius	Where the circle sits and how big	Centre $(-g,-f)$; radius $r=\sqrt{g^2+f^2-c}$	$r>0$; radius is a length	Radius is **not** r^2	Q7(a): centre $(-4,5)$, radius $\sqrt{70}$	—
Exact value	Leave it as a surd, no rounding	e.g. $\sqrt{70}$, not 8.366	"Exact" ⇒ **no decimals**	Not a decimal approximation	Q7(a): radius $=\sqrt{70}$	Wrote **both** $\sqrt{70}$ **and** 8.366 — "exact = no decimals allowed"
Distance between centres	How far apart the two middles are	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$	Straight-line (Pythagoras)	Not the difference of radii	Q7(b): $d=\sqrt{9^2+13^2}=\sqrt{250}=15.81$	—
"Neither touch nor intersect"	The circles miss each other completely	Separated **outside**: $d>r_1+r_2$; or one **inside** other: $d<\lvert r_1-r_2\rvert$	Compare d with r_1+r_2 (and $\lvert r_1-r_2\rvert$)	Not "just say they don't meet" — show the numbers	Q7(b) **[DR]**: sum of radii $\sqrt{70}+\sqrt{52}=15.58<15.81=d$ ⇒ separated	Knew the idea but **stopped** — didn't compute both and state the conclusion in words
Video-ledger conflict	—	—	—	—	Gemini "reconstructed" C_1 radius $=5$	**Corrected:** official radius $\sqrt{70}$ (QP/MS)

1.7 Trigonometry & trig models (Q8)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Pythagorean identity	Turn $\sin^2$ into $\cos^2$ (or back)	$\sin^2 x + \cos^2 x = 1$	All x	Not $\sin x + \cos x = 1$	Q8(i): $5(1-\cos^2 x) + 13\cos x = \cos^2 x$	—
tan identity	tan is sin over cos	$\tan x = \dfrac{\sin x}{\cos x}$	$\cos x \neq 0$	—	Q8(i): clears to $6\cos^2 x - 13\cos x - 5 = 0$	—
Solving in an interval	Find every solution in the allowed range	Get the principal value, then all others in range (e.g. sin: $\theta,\ 180^\circ - \theta$; cos: $\pm\theta$)	Respect the stated interval	Not "just the first answer"	Q8(i): $\cos x = -\frac{1}{3}$ in $0 < x \leq \pi \Rightarrow x = 1.91$ rad; Q8(ii)(a): $\sin(6k+18)^\circ = \frac{5}{6} \Rightarrow 56.4^\circ, 123.6^\circ$	—
Radians vs degrees	Two angle units — the calculator must match the question	π rad $= 180^\circ$	Set mode **per question**	A $(\dots)^\circ$ symbol ⇒ degrees, **not** radians	Q8(i) radians; Q8(ii) model in **degrees**	**Calculator left in radians** for the degree model → wrong $\sin^{-1}(5/6)$
Trig model & "max"	A wave model of temperature; max where $\sin = 1$	$H = 10 + 12\sin(kt+18)^\circ$; max when $\sin = 1 \Rightarrow H = 22$	Use the valid k (here $0 < k < 10 \Rightarrow k = 6.41$)	Not "read off a guess"	Q8(ii)(b): max $H = 22\,$°C; (c): $6.41t + 18 = 90 \Rightarrow t = 11.23$ h ⇒ **11:14**	(b) omitted the °C **unit**; (c) "lost confidence", used 0.985 instead of $k = 6.41$ and stopped at a decimal instead of a clock time
Video-ledger conflict	—	—	—	—	Gemini gave 0.927 from a different equation	**Corrected:** real equation ⇒ $\cos x = -\frac{1}{3}$, $x = 1.91$ — matches the real feedback (Phuc's $1.911 \to 1.91$)

1.8 Differentiation, integration & trapezium rule (Q6, Q9)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Stationary point	Where the curve momentarily flattens	Point where $\dfrac{dy}{dx} = 0$	Solve derivative $= 0$; may be several	Not an intercept	Q9(a): $\frac{dy}{dx} = x^{1/2}(12-5x) = 0 \Rightarrow x = 0$ or $\frac{12}{5}$; P at $x = \frac{12}{5}$	Found y too and treated $x = 0$ as a candidate — **only the x-coordinate of P is asked**; $x = 0$ is the intercept
Power rule (differentiate)	Bring the power down, drop it by one	$\dfrac{d}{dx}x^n = nx^{n-1}$	Any real n	—	$8x^{3/2} - 2x^{5/2} \to 12x^{1/2} - 5x^{3/2}$	—
Power rule (integrate)	Raise the power, divide by the **new** power	$\displaystyle\int x^n\,dx = \frac{x^{n+1}}{n+1} + c$	$n \neq -1$	Not "divide by the old power"	Q9(b): $\int(8x^{3/2} - 2x^{5/2})dx = \frac{16}{5}x^{5/2} - \frac{4}{7}x^{7/2}$	Slipped integrating $x^{5/2}$ (got $\frac{2}{7}x^{7/2}$); correct is $\frac{4}{7}x^{7/2}$

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Signed area	Area below the axis counts as negative	Definite integral is negative below the x-axis	For an *area* below, prepend a **leading minus**	**Not a modulus** — Harry prefers a leading —	Q9(b): R_2 below axis ⇒ leading minus, then $R_1 = R_2 \Rightarrow k^{5/2}(\frac{16}{5} - \frac{4}{7}k) = 0 \Rightarrow k = \frac{28}{5} = 5.6$	Unsure of R_1's upper limit and R_2's sign
Integration variable	Keep x as the thing you integrate; a limit is a different letter	In $\int_a^k f(x)\,dx$, x is the variable, k is the (constant) upper limit	Don't substitute the limit letter into the integrand	The upper limit k is **not** a replacement for x inside f	Q9(b): keep x; k is the end-point, reject $k = 0$ from the diagram ($k > 4$)	Reportedly wrote k in place of x inside the integrand — **[Gemini — source-limited]**
Trapezium rule	Estimate area with straight-line strips	$\int \approx \frac{h}{2}[y_0 + y_n + 2(y_1 + \dots + y_{n-1})]$	Equal strip width h; use **all** tabulated y	Not the exact integral	Q6(b) **[Blank/deferred; MS-only]**: $h = 0.3 \Rightarrow \approx 2.34$ (2 d.p.)	**Left blank in the exam** — this is the homework question
Increasing exponential sketch	$a^x + 4$ climbs and flattens to a line	Curve $y = a^x + 4$ ($a > 1$): through $(0, 5)$, asymptote $y = 4$	Label intercept **and** asymptote	Not a straight line	Q6(a) **[Blank/deferred; MS-only]**: intercept $(0, 5)$, asymptote $y = 4$	Blank in the exam

1.9 Proof & inequalities (Q5)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
"Show detailed reasoning" (**DR**)	Justify **every** non-obvious step in writing	No unexplained jumps; state signs, domains, why steps are valid	Marks lost for missing justification even if the answer is right	Not "just get the answer"	Q5(a) **[DR]**, Q7(b) **[DR]**	See below
Dividing an inequality by a variable	You may divide — but only safely if you know the sign	Dividing by a **positive** quantity keeps the direction; by a negative **flips** it	Must **state** the sign first	Not "always keep the direction silently"	Q5(a): from $3xy(y - x) > 0$ with $x, y > 0$, $xy > 0$ ⇒ direction kept ⇒ $y > x$	**Didn't write** "as x, y are positive, $xy > 0$" → lost a mark on a DR question
Counter-example	One case that breaks a claim	A single instance disproving "for all"	One is enough	Not a general proof	Q5(b) **[Not discussed; MS-only]**: $x = 3, y = -1$ gives $64 > 28$ but $y \not> x$	Part (b) not reached in the lesson

1.10 Exam-technique & notation jargon (whole paper)

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Significant figures (s.f.)	Count from the first non-zero digit	1.911 → **3 s.f.** = 1.91	"unless otherwise stated" default is **3 s.f.**	s.f. ≠ decimal places	Q8(i): 3 s.f. ⇒ 1.91; Q10(a): 3 s.f. ⇒ 2390	Gave 1.911 (4 s.f.); elsewhere rounded to 3 d.p. — a **cross-paper pattern**

Term	Student-friendly meaning	Exact mathematical meaning	Symbols/conditions	What it is NOT	Lesson example	Phuc/common trap
Decimal places (d.p.)	Digits after the point	2.34 → 2 d.p.	Stated per part	d.p. ≠ s.f.	Q6(b): 2 d.p.; Q10(c): 1 d.p.	Confusing 3 s.f. with 3 d.p.
"Show all stages of working" / SW	Calculator-only answers score nothing	"Solutions relying entirely on calculator technology are not acceptable"	Write each line; method marks survive a slip	Not "answer only"	Q2, Q3(i), Q8, Q10(a) marked SW	—
Units	Numbers need their unit	e.g. temperature is $^\circ$C, not a bare number	Include the unit in the final answer	A temperature is **not** an angle	Q8(ii)(b): $22\,^\circ$C	Omitted the °C
"Answer the whole question"	Do the final conversion the question asks for	e.g. "time of day to the nearest minute", not a decimal hour	Read the last line of the part	Not "stop at the decimal"	Q8(ii)(c): $t = 11.23$ h ⇒ **11:14**	Stopped at the decimal hour
Underline-and-tick habit	Underline "3 s.f.", "radians", the unit — then tick each off	A self-check ritual before moving on	—	—	Harry's routine for Q8	Missed precision/mode/unit requirements repeatedly

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2. Cross-tab: Binomial

Method / object	Formula	Use it when...	Conditions / cautions	This paper (May 2024)	Trap
General expansion	$(1+u)^n = \sum_{r\geq 0}\binom{n}{r}u^r$	Expanding a bracket to a power	$n \in \mathbb{Z}^{\geq 0}$ terminates; else need $\lvert u\rvert < 1$	$n = 9$, $u = -\frac{x}{6}$	Sign of u dropped
First four terms	$1 + nu + \frac{n(n-1)}{2!}u^2 + \frac{n(n-1)(n-2)}{3!}u^3$	"first four terms, ascending"	Simplify; order low→high	$1 - \frac{3}{2}x + x^2 - \frac{7}{18}x^3$	Writing $+1x^2$
Coefficient of x^m	Sum of **all** products giving x^m	"coefficient of x^m" (esp. after multiplying by another bracket)	Include every cross-term	$10x \cdot x^2 + 3\cdot(-\frac{7}{18}x^3) \Rightarrow 10 - \frac{7}{6} = \frac{53}{6}$	**Missing the $10x \cdot x^2$ term** (Phuc's lost mark)

3. Cross-tab: Arithmetic & geometric sequences

Method / object	Formula	Use it when...	Conditions	This paper	Trap
Arithmetic nth term	$u_n = a + (n-1)d$	Constant *difference*	$n \in \mathbb{Z}^+$	Q2: $a + 5d = 2$	Using S_n instead
Arithmetic sum	$S_n = \frac{n}{2}(2a + (n-1)d)$	Adding first n terms	—	Q2: $S_{10} = -80 \Rightarrow a = -98, d = 20$	Sum ↔ term mix-up
Smallest n with $S_n > k$	Solve quadratic $S_n - k > 0$	Threshold-crossing	Smallest integer above positive root	Q2(b): $5n^2 - 54n - 4000 > 0 \Rightarrow n = 35$	Slip forming the quadratic (method marks survive)
Geometric nth term	$u_n = ar^{n-1}$	Constant *ratio*	$r \neq 0$	Q10 dormice $r = 1.03$	$n-1$ vs t indexing
Growth after t years	(start) $\times\, (1 + \frac{r}{100})^t$, $t = 0$ at start	"t years after the start"	Exponent is t, not $t-1$	Q10(a): $2000 \times 1.03^6 = 2390$	Used 1.03^{7-1}

Method / object	Formula	Use it when…	Conditions	This paper	Trap
Fit $N = ab^t$ from 2 points	$b^{\Delta t} = \frac{N_2}{N_1}$, then solve for a	Two data points	Answer to stated s.f.	Q10(b) **[MS-only]** $b = 0.98, a \approx 4000$	Reached set-up only
Equate two models	Solve model$_1$ $=$ model$_2$ via logs	“when equal”	Stated precision	Q10(c) **[MS-only]** $T = 13.9$	Not reached

4. Cross-tab: Logarithms

Law / step	Statement	Use it when…	Conditions	This paper	Trap
Definition	$\log_b X = c \iff b^c = X$	Converting log↔index	$b > 0, b \neq 1, X > 0$	Q3	$\log(X+Y) \neq \log X + \log Y$
Power law	$p \log_b X = \log_b X^p$	Coefficient in front of a log	$X > 0$	$2\log_2(2-x) = \log_2(2-x)^2$	Applying to a sum
Combine constant	$c = \log_b b^c$	Turn a number into a log	—	$4 = \log_2 16$	—
Product law	$\log_b X + \log_b Y = \log_b XY$	Adding logs	Same base, args > 0	$4 + \log_2(x+10) = \log_2 16(x+10)$	—
Change of base	$\log_b X = \frac{\log_a X}{\log_a b}$	Awkward base	—	Q3(ii) $\log_{\sqrt{a}} a^6 = 12$	QP base OCR-blurred; MS $= 12$
Domain check	Reject roots with any argument ≤ 0	After solving	—	Q3(i): reject $x = 26$, keep $x = -6$	Keeping the extraneous root (Phuc **got this right** — praised)

5. Cross-tab: Factor / remainder & polynomials

Tool	Statement	Trigger word	Conditions	This paper	Trap
Remainder theorem	Remainder of $f(x) \div (x-a)$ is $f(a)$	“**remainder**”	—	Q4(a): $f(2) = 21$	Doing long division instead
Factor theorem	$(px-q)$ factor $\iff f(\frac{q}{p}) = 0$	“**factor**”	$r = 0$ case	Q4(b): $f(\frac{1}{2}) = 0 \Rightarrow k = 11$	Using $x = \frac{1}{2}$ vs $x = 2$ carelessly
“Hence”	Reuse the earlier result	“**hence**”	—	Q4(c) uses $k = 11$ & factor $(2x-1)$	Restarting from scratch
Fully factorise	Extract known factor, factor the rest	“fully factorise”	Check the quadratic factor	$(2x-1)(x^2+x+1)$	Trying to split an irreducible quadratic
Discriminant	$\Delta = b^2 - 4ac$; sign ⇒ # real roots	“how many real solutions / with a reason”	Real coeffs	Q4(c)(ii): $\Delta = -3 < 0 \Rightarrow$ one real root $x = \frac{1}{2}$	**Not stating** $\Delta < 0$ ⇒ A0

6. Cross-tab: Circle geometry

Tool	Statement	Use it when...	Conditions	This paper	Trap
Centre–radius form	$(x-p)^2+(y-q)^2=r^2$: centre (p,q), radius r	Reading off centre/radius	$r^2>0$	Q7(a): centre $(-4,5)$, $r=\sqrt{70}$	Reporting r^2 as r
Complete the square	$x^2+y^2+2gx+2fy+c=0$	Given expanded form	—	$x^2+y^2+8x-10y=29 \to (x+4)^2+(y-5)^2=70$	Sign errors on g, f
Exact radius	Leave as surd	"exact value"	No decimals	$r=\sqrt{70}$	Writing 8.366 too
Distance between centres	$d=\sqrt{(\Delta x)^2+(\Delta y)^2}$	Comparing two circles	—	$d=\sqrt{250}=15.81$	—
Separation test	Separated iff $d>r_1+r_2$ (or $d<\lvert r_1-r_2\rvert$)	"neither touch nor intersect" **[DR]**	Show both numbers + words	$15.58<15.81$ ⇒ separated	Stopping before the conclusion; **ledger said** $r=5$ **— corrected to** $\sqrt{70}$

7. Cross-tab: Trigonometry

Tool	Statement	Use it when...	Conditions	This paper	Trap
tan $=$ sin / cos	$\tan x=\frac{\sin x}{\cos x}$	Mixed sin tan cos eqn	$\cos x\neq 0$	Q8(i)	—
Pythagorean id.	$\sin^2 x+\cos^2 x=1$	Reduce to one ratio	—	$\to 6\cos^2 x-13\cos x-5=0$	—
Solve as quadratic	Factor / formula in $\cos x$	Quadratic in one ratio	Reject out-of-range root	$\cos x=-\frac{1}{3}$ (reject $\frac{5}{2}$)	—
Interval solutions	sin: $\theta, 180^\circ-\theta$; cos: $\pm\theta$ (+periods)	Range given	Respect interval	$x=1.91$ rad; $56.4^\circ, 123.6^\circ$	Missing the second solution
Mode (deg/rad)	$(\dots)^\circ$ ⇒ degrees; "in radians" ⇒ radians	Every question	Set calculator per part	Q8(i) rad; Q8(ii) deg	**Calc in radians for the degree model**
Model max	$\sin=1$ at max	"maximum value"	Use valid k	max $H=22\,^\circ$C at 11:14	Missing unit / wrong k / stop at decimal

8. Cross-tab: Integration, differentiation & trapezium rule

Tool	Statement	Use it when...	Conditions	This paper	Trap
Differentiate power	$\frac{d}{dx}x^n=nx^{n-1}$	Gradient / stationary pts	—	Q9(a): $12x^{1/2}-5x^{3/2}$	—
Stationary point	$\frac{dy}{dx}=0$	"find the stationary point"	Give only what's asked	P: $x=\frac{12}{5}$	Also solving for y / keeping $x=0$
Integrate power	$\int x^n\,dx=\frac{x^{n+1}}{n+1}+c$	Areas / reverse gradient	$n\neq -1$	$\frac{16}{5}x^{5/2}-\frac{4}{7}x^{7/2}$	Dividing by old power ($\frac{2}{7}x^{7/2}$ slip)
Signed area	Below axis is negative ⇒ leading minus	Region straddling / below axis	**Minus, not modulus**	$R_1=R_2 \Rightarrow k=\frac{28}{5}=5.6$	Sign / limit confusion
Variable vs limit	x is integrated; k is the limit	Definite integral to a variable limit	Don't put k in the integrand	reject $k=0$ from diagram	Wrote k inside integrand — **[Gemini — source-limited]**
Trapezium rule	$\frac{h}{2}[y_0+y_n+2\sum y_{\text{mid}}]$	Estimate a definite integral from a table	Equal h, use **all** y	Q6(b) **[MS-only]** ≈ 2.34	Blank in the exam (homework)

9. Cross-tab: P2 exam technique (Harry's checklist)

Rule	What to do	Where it bit in May 2024	Logged as weakness?
Show all stages (SW)	Write every line; method marks survive a slip	Q2(b): banked M marks despite the quadratic slip	—
Detailed reasoning (DR)	Justify every non-obvious step; state signs	Q5(a) sign of xy; Q7(b) full comparison	**Yes** — inequality justification
Exact means exact	Surds, no decimals	Q7(a): $\sqrt{70}$ only	—
Calculator mode	Degrees vs radians **per question**	Q8(ii): degree model, calc in radians	**Yes** — calculator mode
Rounding	Match s.f. / d.p. exactly (default 3 s.f.)	Q8(i) 1.91; Q10(a) 2390	**Yes** — s.f. not met (cross-paper)
Units	Include °C etc.	Q8(ii)(b): 22 °C	—
Finish the question	Do the final conversion	Q8(ii)(c): 11:14, not 11.23 h	—
Signed area	Leading minus, not modulus	Q9(b)	**Yes** — modulus handling
Sequences routine	Write a few terms; sum vs term vs nth?	Q10(a) t vs $n-1$	**Addressed** this lesson
Command words	remainder→remainder thm; factor→factor thm; hence→reuse	Q4	**Addressed** this lesson
Substitute-back check	If time, plug the answer back	Q2(b) check $S_{35} > 8000$	—

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10. Per-question quick-recall map (status + one-line takeaway)

Wording condensed from the **official QP**; answers from the **MS**; lesson takeaway from the **video**. **DR** = "show detailed reasoning"; **SW** = "solutions relying entirely on calculator technology are not acceptable".

Q (marks)	Official task (QP)	Answer (MS)	Status	One-line takeaway
1(a) — 3	First four terms, ascending, of $(1-\frac{x}{6})^9$	$1-\frac{3}{2}x+x^2-\frac{7}{18}x^3$	Solved	Don't write $+1x^2$. Gemini "$\frac{3}{8}x^2$" note is **[source-limited]**
1(b) — 2	Coeff of x^3 in $(10x+3)(1-\frac{x}{6})^9$	$\frac{53}{6}$	Partly (−1)	Include **every** product giving x^3
2(a) — 4 (SW)	Arith: 6th term $=2$, $S_{10}=-80$; find a, d	$a=-98,\ d=20$	Solved	—
2(b) — 3	Smallest n with $S_n > 8000$	$n=35$	Partly (−1)	Show working → method marks survive the slip
3(i) — 5 (SW)	Solve $2\log_2(2-x)=4+\log_2(x+10)$	$x=-6$	Solved	Reject $x=26$ (domain) — **praised**
3(ii) — 1	$\log_{\sqrt{a}}(a^6)$, $a>1$	12	Solved	Base OCR-blurred; MS confirms 12
4(a) — 1	Remainder of $f(x) \div (x-2)$	21	Taught	"remainder" ⇒ remainder theorem $f(2)$
4(b) — 2	$(2x-1)$ factor ⇒ show $k=11$	$k=11$	Solved	"factor" ⇒ $f(\frac{1}{2})=0$
4(c)(i) — (5)	Fully factorise $f(x)$	$(2x-1)(x^2+x+1)$	Solved	Discriminant as a "does it factorise?" check
4(c)(ii) —	# real solutions, with a reason	one ($x=\frac{1}{2}$)	Solved	**State** $b^2-4ac=-3<0$
5(a) — 4 (DR)	$x, y>0$, $(x-y)^3 > x^3-y^3$; prove $y>x$	$y>x$	Solved (fix)	Write "as $x, y>0$, $xy>0$" before dividing

Q (marks)	Official task (QP)	Answer (MS)	Status	One-line takeaway
5(b) — 2	Counter-example for all reals	e.g. $x = 3, y = -1$	**Not discussed** [MS-only]	—
6(a) — 3	Sketch $y = a^x + 4$ $(a > 1)$: intercept & asymptote	$(0, 5)$; $y = 4$	**Blank/deferred** [MS-only]	Homework question
6(b) — 3	Trapezium rule for $\int_2^{3.5} \frac{2^x - 2x}{x}\, dx$-type	≈ 2.34	**Blank/deferred** [MS-only]	Use all tabulated y
6(c) — 3	Estimate (i) & (ii) using (b)	18.84; 4.68	**Blank/deferred** [MS-only]	—
7(a) — 3	Centre & exact radius of $C_1 : x^2 + y^2 + 8x - 10y = 29$	$(-4, 5)$; $\sqrt{70}$	Solved (fix)	Exact = no decimals; ledger "$r = 5$" **corrected**
7(b) — 3 (DR)	Prove C_1, C_2 neither touch nor intersect	$15.58 < 15.81$	Solved (fix)	Compute both, conclude in words
8(i) — 5 (SW)	Solve $5 \sin x \tan x + 13 = \cos x$, $0 < x \leq \pi$, rad 3 s.f.	$x = 1.91$	Solved (fix)	4 s.f. → 3 s.f.; ledger "0.927" **corrected**
8(ii)(a) — 4	$H = 10 + 12 \sin(kt + 18)^\circ$, $H = 20$ at 6 am; all k	$k = 6.41, 17.59$	Solved (fix)	Degree model — set calc to degrees
8(ii)(b) — 1	Max temperature $(0 < k < 10)$	$22\,^\circ$C	Solved (fix)	Include the °C
8(ii)(c) — 2	Time of that max, nearest minute	11:14	Solved	Finish: convert 11.23 h → 11:14
9(a) — 4	x-coord of stationary P of $y = 2x^{3/2}(4 - x)$	$x = \frac{12}{5}$	Solved	Only x; $x = 0$ is the intercept
9(b) — 4	$R_1 = R_2$; exact k	$k = \frac{28}{5} = 5.6$	Solved	Leading minus for R_2; $\int x^{5/2} = \frac{2}{7}x^{7/2}$ slip; "k in integrand" **[source-limited]**
10(a) — 2 (SW)	Dormice after 6 yr, 3 s.f.	2390	Solved	Exponent t, not $t - 1$; 3 s.f. not 3 d.p.
10(b) — 3	Voles $N = ab^t$; a, b to 2 s.f.	$a \approx 4000,\ b \approx 0.98$	**Setup only** [MS-only]	Ledger "$A + B(1.2)^t$" **corrected** to geometric
10(c) — 3	When dormice $=$ voles, T to 1 d.p.	$T = 13.9$	**Not reached** [MS-only]	—

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11. Conflicts resolved (Gemini video ledger → official QP/MS)

Every place the video reconstruction disagreed with the paper, **the QP/MS were used**:

Q	Gemini ledger said	Official QP/MS (used here)
2	6th term -5; $S_n > 2000$	6th term 2; $S_n > 8000$; $a = -98, d = 20, n = 35$
3	$\log_x$ forms; $\log_2(x^2)$	$2\log_2(2 - x) = 4 + \log_2(x + 10) \Rightarrow x = -6$; $\log_{\sqrt{a}}(a^6) = 12$
7	C_1 radius $= 5$	radius $\sqrt{70}$; sum $15.58 <$ distance $\sqrt{250} = 15.81$
8(i)	answer 0.927 (different eqn)	$\cos x = -\frac{1}{3} \Rightarrow x = 1.91$ rad — *matches* the real feedback
9(a)	"coordinates of P" incl. y	only the x-coordinate is asked; $x = \frac{12}{5}$
10(b)	$N = A + B(1.2)^t$	geometric $N = ab^t$, $a \approx 4000,\ b \approx 0.98$

Genuinely unresolved (no lesson evidence): Q5(b), all of Q6, Q10(b)-full and Q10(c) — answers here are **MS reference only**. **Source-limited Gemini reconstructions:** G1 (Q1a "$\frac{3}{8}x^2$") and G2 (Q9b "k in the integrand") — flagged, not asserted.

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12. Homework & priority weaknesses (study log, 10 Jul 2026)

- **Finish Q6** (curve sketch + trapezium rule), left blank → then **email Harry** for the aggregated score/grade. Also finish Q10(b)-full and Q10(c).
- Revisit **P3 Oct 2025**; complete **P3 Jun 2024** before Sunday's tutorial (textbook allowed, no time pressure).
- **Book more tutorials** for the coming week (only Sun 19 Jul booked).

Four weaknesses logged this lesson: (1) inequality justification (Q5), (2) calculator mode (Q8), (3) significant figures not met (Q8/Q10, cross-paper), (4) modulus handling (prefer leading minus). **Two marked addressed:** geometric "sum vs nth term" confusion, and remainder-theorem confusion.

Grounded in `sources/` (official QP + MS, study log) and `analysis/` (Gemini video evidence), reconciled against the final passed guide. No question, mark, value or tutor comment has been invented. MS-only and Gemini source-limited items are labelled inline. Not a release — internal recall aid.